GLOBAL COMMODITY TRADING ENGINE

An Open, Federated and Future-Adaptable Infrastructure for Global Commodity Markets

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Conceptual proposal; not a representation that the Engine has been implemented, certified, commercially validated or approved as a regulated trading venue. References to standards bodies, governments and industry initiatives document external facts; they do not imply endorsement, affiliation or political alignment.

Abstract

Global commodity trade comprises diverse vertical markets—agricultural products, timber, energy, ores, metals and many others—served by specialized intermediaries, financial institutions, inspectors and logistics operators. Digitalization has progressed, but incompatible systems, repeated integrations, fragmented trust arrangements and dependence on proprietary platforms continue to constrain interoperability. SFOC articulates a technology-neutral **Global Commodity Trading Engine (GCTE)** concept: a federated infrastructure framework that makes common trade capabilities reusable across independent commodity verticals without creating a mandatory central exchange, new currency or competing inspection institution. The concept combines open interfaces, verifiable organizational identity, trusted evidence exchange, independent governance, provider substitutability and cryptographic agility. Existing specialist providers provide evidence of demand for digital commodity services, while failures of large industry-sponsored platforms expose adoption and governance risks. India is a strategically important reference case for interoperable digital infrastructure and commodity-market modernization; its existing initiatives offer useful evidence, not proof of GCTE feasibility. Independent future implementers may explore Indian and other emerging-market trade corridors alongside opportunities worldwide, on commercial and technical merits rather than political affiliation. Its feasibility, legal applicability, commercial incentives and security remain unproven and would require staged, measurable trials by independent implementers. Its conceptual contribution is a **testable institutional and technical architecture hypothesis**, not a claim to have invented interoperability or blockchain.

Keywords: commodity trade; interoperability; federation; distributed governance; verifiable credentials; electronic trade documents; post-quantum cryptography; crypto agility; modular infrastructure.

1. Purpose and central proposition

The objective is not to sell a proprietary trading platform, replace established professional service providers, or force all markets onto one ledger. It is to test whether independent commodity markets can

share a durable set of interoperable infrastructure capabilities while retaining their commercial autonomy, specialized rules and freedom to choose service providers.

Core proposition: common infrastructure should be shared; market relationships, commercial decisions and legitimate regulatory responsibilities should remain distributed.

The GCTE would enable direct bilateral trading, industry-specific marketplaces and established trading venues to interoperate voluntarily. It would not itself be the mandatory counterparty, price setter, principal custodian of commodities or exclusive gatekeeper to global trade. Independent operators should be able to continue trading if any one technology supplier withdraws.

SFOC publishes the vision for critical technical, commercial, legal and academic examination; it does not undertake to develop, fund, own, operate or control an Engine or its markets. All architectural choices below are proposed design criteria or testable hypotheses unless identified as existing external standards.

1.1 Institutional boundary: SFOC does not develop, finance or control the Engine

SFOC serves as an independent catalyst for new ideas, constructive professional dialogue and international networking. Its role is to articulate, attribute and publish the vision, invite critical perspectives and connect expertise—not to direct an implementation. SFOC is **not** the prospective software developer, project financier, exchange operator, validator authority, commercial intermediary, data custodian or controller of a future GCTE network. Neither publication of this paper nor use of the term “Engine” grants SFOC any operational, economic or governance authority over markets, counterparties or an eventual implementation.

If independent companies, institutions or consortia later decide to build or adopt compatible solutions, **they alone** must arrange development, capital, regulatory permissions, governance, operation, liability and ongoing support. Any such implementation must be governed so that no single participant—including an initial technology partner or funder—can acquire exclusive market control. Intellectual attribution of the SFOC vision is distinct from ownership or governance of future implementations.

2. The problem: fragmentation rather than absence of technology

The International Chamber of Commerce's Digital Standards Initiative (ICC DSI) stated on 2 October 2026 that accelerating fragmentation in digital trade requires a renewed focus on standards, trust, identity and interoperability at scale [1]. This is a direct external reference for the problem addressed here, **not evidence that the GCTE has been validated or endorsed**.

The market has already achieved material progress. A June 2026 Digital Container Shipping Association (DCSA) announcement described cross-platform electronic bill of lading interoperability among five providers [2]. The FIT Alliance's October 2026 survey identified interoperability as a defining challenge of further adoption [3]. Accordingly, the GCTE must complement rather than duplicate existing eBL and trade-document initiatives.

2.1 Market validation through existing providers

The presence of established specialist solutions is evidence of commercial demand for parts of the digital-trade value chain, not proof of demand for the entire proposed Engine. **Covantis** is an operational post-trade platform serving agricultural commodities [11]; **VAKT** provides digital workflows for physical commodity post-trade, particularly energy markets [12]; and **Komgo** connects corporates and financial

institutions for trade-finance operations [13]. In June 2026, DCSA announced five eBL platform providers adopting its interoperability annex—CargoX, edoxOnline, TradeGo, WaveBL and eTEU [2]. These are not interchangeable products, and some may be future integration partners rather than direct substitutes.

Market implication: SFOC does not assume a greenfield market. The Engine must show incremental value beyond existing vertical applications and multi-platform standards, such as lower repeated integration cost, independent governance, provider portability and reusable multi-vertical workflows. Commercial validation must test these advantages explicitly.

2.2 TradeLens and the independent-entry hypothesis

A cautionary example is **TradeLens**, initiated by Maersk and IBM. In November 2022, its founders announced discontinuation despite having developed a viable platform, citing insufficient global industry collaboration and commercial viability [4]. This makes TradeLens a documented example of a technically viable solution failing to secure necessary industry-wide participation; it does **not** establish that incumbent ownership alone caused failure.

An independently organized implementation could test a contrasting **lean, corridor-first ("guerrilla") entry strategy**: no incumbent-controlled central exchange, no mandatory network lock-in, demonstrable participant-level benefits, transparent governance from the outset and credible technical exit rights. It is a hypothesis that a small, supplier-neutral initiative can reduce perceived conflicts of interest; it is not a claim that outside initiatives necessarily succeed or established firms cannot cooperate.

Commodity markets also have sector-specific friction: inconsistent quality descriptions, origin or sustainability evidence, financing documentation, warehouse records, logistics updates, ownership transfer rules and dispute procedures. A universal application that ignores those differences would simply reproduce a new centralized bottleneck.

3. Non-negotiable design principles

1. **No mandatory central exchange.** Price discovery, brokerage and exchange services remain competitive and optional; the Engine is connective infrastructure, not a new exclusive venue.
2. **No proprietary currency or payment instrument.** Payments occur in established sovereign currencies through regulated and legally accepted financial rails. The Engine may coordinate payment evidence and workflows but does not itself become a bank or issuer.
3. **No competing inspection authority.** Existing qualified inspectors, certifiers, laboratories and surveyors remain responsible for physical assessments. The Engine verifies provenance and integrity of their signed attestations, not the physical truth of their observations.
4. **No single controlling technology vendor.** Source availability, interface specifications, portability, continuity plans and governance must support exit and replacement of providers.
5. **Plug-in / plug-out by design.** Providers, identity systems, cryptographic components, document standards and applicable workflow modules must be replaceable subject to certification, compatibility and legal constraints.

6. **Privacy and commercial autonomy.** Sensitive quantities, prices, counterparties, strategies and contracts are not automatically published to a shared public ledger. Access remains controlled by legitimate parties and applicable law.
7. **Standards first, technology second.** Blockchain, permissioned ledgers, registries, distributed consensus and other architectures are options evaluated by function, not ideological requirements.
8. **Crypto agility and post-quantum readiness.** Security components must support controlled algorithm migration and durable verification; quantum-resistant operation is not presumed to exist at the outset.
9. **Independent, accountable governance.** Decisions and rules must be explicit, auditable, contestable and resistant to capture by one market participant or supplier.
10. **Incremental adoption with measurable value.** Each vertical must qualify its business case before expansion; there is no compulsory global migration.

4. Reference architecture: one framework, many independent verticals

The proposed design has four separable layers. This is a logical architecture, **not** a decision to deploy a single global blockchain.

4.1 Vertical layer — commodity-specific rules

A commodity vertical supplies domain-specific definitions of products, grades, measurement units, permitted documents, inspection methods, delivery terms, applicable law, exception handling and market conventions. A timber transaction and an energy transaction need not use the same commercial workflow merely because they share identity and verification primitives. Existing exchanges and bilateral networks can integrate without surrendering their market governance.

4.2 Common Engine capabilities — reusable, not monopolistic

Potential common capabilities include organizational identity and role verification; standardized event and document schemas; a directory of qualified services; signed document and inspection attestations; permissioned evidence exchange; workflow orchestration; cross-platform state reconciliation; audit trails; consent and access policies; and structured dispute hand-offs. Only capabilities whose cross-vertical value is demonstrated should become shared modules.

4.3 Provider adapter layer — regulated and professional institutions

Payment, banking, trade finance, inspection, insurance, logistics, customs and data providers connect through versioned interfaces. The Engine does not replace their mandates. A provider's removal must trigger controlled reassignment, preservation of records and continuity of in-flight transactions. **Plug-out** is an acceptance test, not merely a contractual aspiration.

4.4 Federation and evidence layer — independent operation

Independent nodes or services can be operated by mutually independent organizations. The smallest viable shared state should be chosen: signatures, timestamps, revocation status and integrity proofs

where helpful; confidential business content remains off-ledger or within authorized domains. A common ledger should be used only where its added value over signed records and interoperable registries is demonstrated.

5. Identity, trust and the physical world's first mile

Digital signatures can establish who asserted a fact and whether a record changed; they cannot establish that a cargo's physical quality, mass or origin was correctly measured. The Engine therefore separates **source evidence**, **issuer authority**, **digital integrity** and **legal responsibility**.

Organization identity and delegated authority should be verified using appropriate recognized schemes. The verifiable Legal Entity Identifier (vLEI), maintained in the Global Legal Entity Identifier Foundation ecosystem, is one relevant option—not an exclusive requirement [5]. Signatures, credentials and issuer registries must support expiry, revocation, dispute, audit and re-verification.

A minimum inspection attestation would identify the issuing organization and authorized signatory, lot or shipment identifier, scope, methods, date, measured results and applicable accreditation, together with a verifiable signature and integrity reference. Actual certificates may remain private. Independent laboratories and inspectors retain responsibility for their findings.

First-mile data controls require source attribution, calibrated measurements where relevant, chain-of-custody practices, dual confirmation for high-risk events and explicit uncertainty handling. False input must remain discoverable and disputable; immutable recording must not convert an error into unquestionable truth.

6. Legal validity, title, payments and dispute resolution

A record hash does not by itself transfer ownership of a physical commodity or establish the legal effectiveness of an electronic bill of lading. The UNCITRAL Model Law on Electronic Transferable Records (MLETR) sets technology-neutral principles for electronic transferable records, including identification, integrity and exclusive control; the applicable national laws and transaction agreements remain decisive [6]. The Engine must map which instruments are legally recognized in each relevant jurisdiction and preserve paper or alternative procedures where necessary.

Payment execution remains with authorized institutions. The Engine may coordinate documentary conditions, payment status notifications and release instructions, but must not promise atomic settlement or elimination of counterparty risk where banking rails, contracts and law do not provide it.

Dispute resolution must have explicit governing law, jurisdiction or arbitration clauses, evidence preservation, emergency procedures and the ability to correct or supersede erroneous records through transparent authorized processes. Sanctions, AML/KYC and applicable market rules must be enforced by the actors legally responsible for them; decentralization is not an exemption from compliance.

7. Minimum viable governance: decentralization with accountability

“Decentralized governance” is not achieved merely by distributing servers. Independent implementers could test a multi-stakeholder governance model with separately represented constituencies—for example trading participants, independent service providers, technology operators and independent

public-interest or standards expertise. Exact legal form is deferred until participating jurisdictions and founding members are known.

Initial governance safeguards should include: published eligibility and voting rules; limits on concentration of voting influence; conflict-of-interest disclosure and recusal; transparent proposal, consultation and supermajority processes for material protocol changes; minority appeal rights; independent security review; documented incident powers with expiry and review; and clear intellectual-property and exit provisions. A participant should not gain more governance power solely by holding a proprietary token.

The governing entity, if any, should be **steward of a protocol**, not owner of customer trades. The protocol operator, validator operators, auditors, software developers and commercial venues should be separately accountable wherever feasible. Vendor-switch and network-continuity exercises should be conducted before a production declaration. Governance costs and slow decision-making are explicit risks to test, not overlooked inconveniences.

8. Crypto agility and post-quantum transition

The June 2026 update to NIST's *Considerations for Achieving Crypto Agility* describes approaches to adapting and replacing cryptographic algorithms while preserving security and operations [7]. NIST standards FIPS 203 (ML-KEM), FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) supply relevant standardized post-quantum building blocks [8]. These are references for evaluation, not a claim that current identities, networks, financial signatures or the proposed Engine are already quantum-safe.

A realistic security design requires a cryptographic inventory; modular key management and trust stores; algorithm and protocol versioning; hybrid or staged migration where appropriate; revocation and re-signing procedures; long-term archive verification; and interoperability testing with existing institutions. The impact of signature size, latency, standards compatibility and legal acceptance must be measured. Future procedures must support replacing compromised algorithms and vendors without rewriting transaction history or sacrificing evidentiary continuity.

Terminology note: “lattice-based cryptography” is a mathematical family of cryptographic techniques; it is not interchangeable with a distributed organizational “lattice” or network topology.

9. Adoption incentives and viable operating economics

Openness does not create adoption by itself. The Engine must give each participant a reason to join without surrendering commercially valuable relationships. A first adopter should gain a demonstrable reduction in duplicated data entry, repeated onboarding, document reconciliation, administrative latency or counterparty-verification effort. Independent service providers should retain customer ownership and freedom to charge for their actual services.

A possible non-exclusive operating model combines transparent, cost-reflective infrastructure fees; paid implementation and support services offered competitively by multiple suppliers; and optional commercially operated vertical applications. Any governance body must be financially sustainable without relying on token speculation, mandatory transaction intermediation, exclusive data brokerage or one dominant sponsor. Fee fairness, cost allocation and possible free-riding require explicit design and testing.

Investability should be assessed separately from public-protocol stewardship. Commercial opportunities could exist in connectors, integration, hosting, verification, vertical software and support, while the shared protocol remains independently governed. No market-size, return-on-investment or revenue forecast is asserted in this concept stage.

9.1 Emerging-market corridors and investor diversity: commercially grounded, politically non-aligned

For potential independent implementation, **India and selected emerging-market trade corridors in Asia and Africa warrant focused commercial and technical examination**, while participation and funding remain open to all regions. SFOC identifies this opportunity but neither selects or finances pilots nor operates any resulting infrastructure. Candidate corridors are to be compared on transaction friction, documented demand, willing independent counterparties, legal feasibility, governance safeguards and measurable commercial benefits. This is a geographic and economic research hypothesis, **not an endorsement of any government, political grouping or trade bloc**, and not a policy of excluding established institutions from Europe, North America or elsewhere.

Independent public sources indicate practical needs across several regions: the **Asian Development Bank's 2025 Global Trade Finance Gap Survey** estimates a worldwide trade-finance gap of USD 2.5 trillion [15], while the **African Union's AfCFTA Protocol on Digital Trade** calls for open standards and cross-border interoperability [16]. The **BRICS New Delhi Declaration of September 2026** also mentions digital public infrastructure, trade-document digitization and access to trade finance for smaller enterprises [14]. It is cited **only as documentary evidence of stated economic and technical priorities in certain markets**, not as an expression of support for any political agenda or as an affiliation with BRICS. None of these sources demonstrates GCTE demand, investor commitment, product feasibility or SFOC endorsement by the issuing institutions.

Prospective funding sources could include independently governed industry pilot consortia, commercial strategic investors, regional venture and infrastructure funds, development-oriented institutions and research grants, each subject to actual investment mandates and eligibility. **No investor may buy exclusive protocol control, privileged access to other parties' trade data or permanent governance dominance.** Any development-finance participation would require separate qualification; no institution is claimed as a current partner or prospective funder by name.

9.2 India: a strategic reference case, not a political alignment

India deserves substantive attention because its real-world commodity diversity, large network of trading participants and experience with modular digital public infrastructure create unusually relevant conditions for examining interoperability. Its importance to this paper is economic, operational and architectural—not geopolitical. The objective is neither to propose an Indian national exchange nor to displace India's established institutions, but to understand how independently operated markets could connect across existing systems while preserving local rules, counterparties, privacy and service-provider choice.

India provides tangible precedents. The Government of India's **National Agriculture Market (e-NAM)** reported **1,656 integrated mandis across 23 states and four union territories by March 2026**, demonstrating the scale of digitization of agricultural market relationships [17]. The **Trade Connect e-Platform**, launched in September 2024, supports information and connections for exporters, including smaller businesses [18]. India's G20 work on **Digital Public Infrastructure (DPI)** promoted minimal, reusable, interoperable digital building blocks as a cross-sector design approach [19]. The Reserve Bank of

India has also identified opportunities for DPI to support cross-border commerce and sectors including agriculture and MSMEs [20]. These systems have different purposes, governance and legal scopes; **none is presented here as a GCTE prototype, technical sponsor or proof that decentralized global commodity settlement already exists**.

For prospective independent technical and commercial participants, India therefore offers a **priority research and dialogue perspective** spanning commodity-specific workflow integration, authenticated inspection evidence, cross-border documentation, trade finance and vendor-independent connectors. Possible India-linked trade corridors should be assessed with actual counterparties, under Indian and other applicable laws, and compared transparently with candidates elsewhere. An Indian technology developer, commodity trader, bank or investor might independently participate in a future initiative, but no participation or financing is assumed. **SFOC neither develops, finances nor controls such activity.**

10. Pilot selection and evidence-based migration

Selection of a first commodity vertical and trade corridor should follow comparative assessment, not personal preference or geopolitical assumptions. Criteria include standardized product definitions, repeatable transaction flows, willing independent counterparties, identifiable document friction, legally feasible digitization, availability of trusted data issuers, practical benefits for producers and smaller traders, and a credible route to commercial sustainability. Selected emerging-market corridors merit examination but must compete under the same objective criteria as candidates in every other region.

A proposed **12–18 month research and pilot horizon** is an indicative planning hypothesis, not a delivery commitment:

- **Phase A — Requirements and baseline:** map one candidate transaction lifecycle, legal constraints, affected actors, incumbent systems, current costs and failure modes.
- **Phase B — Interoperability demonstrator:** implement one end-to-end multi-provider process using recognized document and identity standards, with no migration of real financial custody.
- **Phase C — Controlled live pilot:** with consenting regulated participants, execute a narrow legally reviewed workflow under independent security and operational oversight.
- **Phase D — Replication and exit test:** demonstrate a second independent operator or vertical and remove one service provider without loss of records, control or continuity.

Success should be judged against matched current-process baselines. Suggested measurements are: total document handling time; avoidable manual touchpoints; onboarding effort; reconciliation discrepancies; participant concentration; provider-switch recovery time; security incidents; all-in cost per completed transaction; and number of genuinely independent participants.

11. Testable hypotheses and research agenda

H1 — Interoperability: A standards-based Engine demonstrator can complete a comparable multi-party document workflow with less duplicated integration or reconciliation effort than the selected baseline.

H2 — Provider independence: A certified provider can be replaced during an agreed test scenario without loss of legally relevant evidence or control over active workflows.

H3 — Commercial adoption: At least two independent market-side organizations perceive sufficient net benefit to participate without exclusivity or mandatory proprietary-currency requirements.

H4 — Trust: Qualified counterparties can verify the issuer, scope, integrity and revocation state of selected third-party attestations without consulting a single proprietary gatekeeper.

H5 — Security evolution: A cryptographic algorithm transition can be executed in a controlled test environment while preserving verifiability of historical evidence and availability of the selected workflow.

H6 — Independent entry and corridor value: At least one selected corridor demonstrates measurable participant benefit and independent multi-party adoption without requiring a dominant incumbent, proprietary currency or compulsory exclusive platform.

For scientific assessment, each hypothesis needs a preregistered test protocol, explicit baseline, sample definition, failure criteria and reproducible results. Hypotheses may fail; a scientifically credible result records that failure rather than redesigning the claim after observing outcomes.

12. Principal risks, assumptions and decision gates

- **Governance capture:** independent constituencies may be dominated through funding or informal influence. *Gate:* enforceable concentration and recusal safeguards.
- **Commercial disincentive:** incumbents may prefer closed networks. *Gate:* signed pilot participation and validated economic advantage rather than stated enthusiasm.
- **Legal incompatibility:** electronic instruments may lack recognition across a trade corridor. *Gate:* instrument-by-instrument and jurisdiction-by-jurisdiction legal qualification.
- **First-mile error or fraud:** signed evidence may still contain false measurements. *Gate:* credible issuer accountability, source controls and dispute procedures.
- **Privacy leakage:** a shared record may reveal pricing, sources or commercial relationships. *Gate:* tested data-minimization and access-control design.
- **Vendor lock-in:** nominally open interfaces may hide proprietary dependencies. *Gate:* successful provider replacement and open-format export tests.
- **Cybersecurity and PQC transition:** insecure implementations or premature algorithm claims may create new risk. *Gate:* threat modeling, independent assessment and measured migration testing.
- **Unsustainable economics:** a network may work technically without funding broad participation. *Gate:* documented costs, funding sources and credible break-even hypotheses.
- **Funding-related capture:** an investor or regional bloc could seek influence over access, fees, code or governance. *Gate:* enforceable ownership, voting, data-access and conflict-of-interest safeguards across all funding sources.

- **Corridor and sanctions exposure:** trade routes may involve legal, financial, sanctions or export-control complexity across jurisdictions. *Gate:* transaction-specific legal screening and compliance responsibilities, independent of geography or funding origin.

These are unresolved risks. The present paper does not establish commercial, legal or technical feasibility.

13. Relationship to existing initiatives and differentiation

The GCTE is conceived as an **implementation and coordination architecture that should reuse existing standards**, not replace their organizations. ICC DSI and its Key Trade Documents and Data Elements (KTDDE) efforts address standardization and trade-data harmonization [1, 9]. DCSA already specifies eBL interoperability patterns [2]. UNCITRAL MLETR provides a technology-neutral legal model [6]. vLEI offers one organizational trust mechanism [5]. NIST provides crypto-agility and post-quantum reference material [7, 8].

Commercial competitors such as Covantis, VAKT and Komgo demonstrate functioning sector-specific markets, while DCSA demonstrates real eBL cross-platform interoperability [2, 11–13]. Any future implementation would need to demonstrate complementarity and added value rather than imply these capabilities do not exist.

The proposed distinct contribution is the **combination to be tested**: vertical-independent reusable trade capabilities; decentralized but accountable protocol governance; explicit provider plug-out; integration of existing payment and inspection services; confidentiality-preserving federation; and a quantified, multi-vertical migration method. This combination is a research and design hypothesis, not an established patentable invention or verified market novelty. A formal prior-art and standards gap analysis is required before stronger originality claims.

14. Invitation to independent examination

SFOC makes this final concept available for independent examination by commodity participants, inspectors, banks, insurers, logistics operators, technology developers, researchers and prospective investors, including interested organizations in India, other Asian and African markets, Europe and the Americas. Any decision to develop, finance, govern or operate an implementation belongs exclusively to independently organized third parties; no SFOC funding or operational participation is implied. No supplier has a privileged technical position merely by possessing an existing blockchain product. A future technical partner would be judged by conformance, independence, security, portability and the demonstrated value of its contribution.

The goal is not to substitute one centralized gatekeeper for another. It is to examine whether independent commodity markets can share trustworthy infrastructure without giving up control of their own trade.

Scope and scientific status

This is a **final, independently published conceptual white paper**, not peer-reviewed research. A separate final conceptual research article develops the questions, architecture propositions and evaluation design more systematically. Neither this paper nor that article presents results of a deployed

Engine, field trials or independent peer review. Any empirical article would require documented tests and reproducible comparative findings. Institutional attribution identifies SFOC as publisher and originator of the stated concept; it does not represent an exclusive claim to pre-existing standards, techniques or unimplemented functional ideas. Individual authorship and rights-transfer questions, where legally relevant, remain governed by applicable law.

Related scientific publication

The companion scientific concept paper was published first on 3 November 2026. It develops the proposed architecture, research hypotheses, evaluation method and limitations in greater detail. It is an independently accessible preprint, not a peer-reviewed or empirically validated study. See reference [21]. https://doi.org/10.5281/zenodo.23912911

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